

Introduction to Real Analysis – Homework 3

Lebesgue measurability and approximation

Notation. m^* denotes Lebesgue outer measure. All sets lie in $\mathbb{R}^d$ unless stated otherwise.

1. **Sets with null boundary.** Write E° for the interior of E and $\partial E = \overline{E} \setminus E^\circ$ for its boundary.

(a) If $m^*(\partial E) = 0$, prove that E is Lebesgue measurable and

$$m^*(E^\circ) = m^*(E) = m^*(\overline{E}).$$

(b) Give a Lebesgue measurable subset of $\mathbb{R}$ whose boundary has positive outer measure.

2. **Approximation from both sides.**

(a) If E is Lebesgue measurable, prove that for every $\varepsilon > 0$ there is a closed $F \subset E$ with $m^*(E \setminus F) < \varepsilon$. *Hint:* apply the definition of measurability to E^c .

(b) Prove that E is Lebesgue measurable if and only if, for every $\varepsilon > 0$, there exist a closed F and an open U such that

$$F \subset E \subset U, \quad m^*(U \setminus F) < \varepsilon.$$

The set E need not be bounded or have finite outer measure.

3. **A countable intersection of open sets.** A G_δ set is a countable intersection of open sets. Prove that E is Lebesgue measurable if and only if

$$E = G \setminus N$$

for some G_δ set G and some $N \subset G$ with $m^*(N) = 0$. *Hint:* choose open supersets whose excess outer measures tend to zero.

4. **Challenge: compact approximation without boundedness.**

(a) If $A \subset \mathbb{R}^d$ and $m^*(A) < \infty$, prove that for every $\varepsilon > 0$ there exists $R > 0$ such that

$$m^*(A \setminus [-R, R]^d) < \varepsilon.$$

Hint: take one box cover of finite total volume and use the tail of that series.

(b) If E is Lebesgue measurable and $m^*(E) < \infty$, prove that for every $\varepsilon > 0$ there is a compact set $K \subset E$ with

$$m^*(E \setminus K) < \varepsilon.$$

Use part (a) and Problem 2(a).